

Generative Adversarial Networks (GANs)

Generative models: output objects of
real world, e.g. images, texts,
molecules, etc.

$\{x_1, x_2, \dots, x_n\}$ - dataset, e.g. MNIST image

Goal: $x \sim p(x)$

$$GAN: \quad z \in \mathbb{R}^d, \quad z \sim p(z) = \mathcal{N}(z | 0, I)$$

$$x = g(z, \theta) - \text{generator}$$

$$p(x|\theta) \quad \text{log-likelihood loss:}$$

$$\text{delta function} \quad -\frac{1}{N} \sum_{i=1}^N \log p(x_i | \theta) \rightarrow \text{nw} \quad \theta$$

$$p(x|\theta) = \int \delta(x - g(z, \theta)) p(z) dz$$

- can't be computed

$$\delta(x-a) = \begin{cases} 0, & \text{if } x \neq a \\ +\infty, & \text{if } x = a \end{cases}$$

$$\int_{-\infty}^{+\infty} \delta(x-a) dx = 1$$

$$\begin{aligned} \delta[f] &= \\ &= \int \delta(x-a) f(x) dx = \\ &= f(a) \end{aligned}$$

discriminator $d(x) \begin{cases} \rightarrow \text{"real"} \\ \rightarrow \text{"fake"} \end{cases}$ $d(x) \in (0, 1)$
 (add σ -function to the last layer)

fixed g : $\mathbb{E}_{x \sim p_{\text{data}}(x)} \log d(x) + \mathbb{E}_{x \sim p_{\text{gen}}(x|\theta)} \log(1-d(x)) \rightarrow \max_d$

fixed d : $\underbrace{\mathbb{E}_{x \sim p_{\text{data}}(x)} \log d(x)}_{\text{const}} + \mathbb{E}_{p(z)} \log(1-d(g(z, \theta))) \rightarrow \min_g$

GAN scheme

$$g(z, \theta_g), d(x, \theta_d)$$

Init. θ_g, θ_d

Until convergence:

may be
several
times

$$x_1, x_2, \dots, x_M \sim p_{\text{data}}(x)$$

$$z_1, z_2, \dots, z_M \sim p(z) = \mathcal{N}(z|0, I)$$

$$\mathcal{L}_d = -\frac{1}{M} \sum_{j=1}^M \left(\log d(x_j, \theta_d) + \log (1 - d(g(z_j, \theta_g), \theta_d)) \right)$$

$$\theta_d = \text{opt_step}(\nabla \mathcal{L}_d)$$

$$z_1, z_2, \dots, z_M \sim p(z)$$

$$\mathcal{L}_g = \frac{1}{M} \sum_{j=1}^M \log(1 - d(g(z_j, \theta_g), \theta_d))$$

$$\theta_g = \text{opt_step}(\nabla \mathcal{L}_g)$$

Fixed g : $\int p(x) \log d(x) dx + \int q(x) \log (1-d(x)) dx$

$\nwarrow$ data distr. $\nwarrow$ gen. distr.

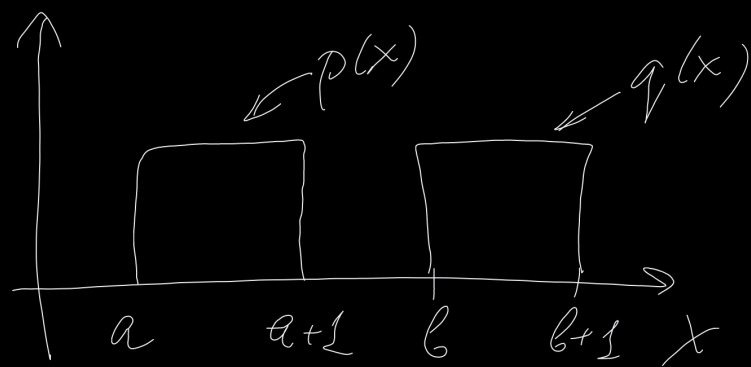
Fix x : $p \log d + q \log (1-d) \rightarrow \max_{d \in (0,1)} \rightarrow \max_d$

$$\frac{\partial}{\partial d} = \frac{p}{d} - \frac{q}{1-d} = 0 \Rightarrow p - pd - qd = 0 \Rightarrow$$

$$\Rightarrow d = \frac{p}{p+q}$$

$$d_{\text{opt}}(x) = \frac{p(x)}{p(x) + q(x)}$$

$$\begin{aligned}
& \int p(x) \log \frac{p(x)}{\frac{1}{2}(p(x)+q(x))} dx + \int q(x) \log \frac{q(x)}{\frac{1}{2}(p(x)+q(x))} dx + \\
& + 2 \log \frac{1}{2} = \text{KL} \left(p(x) \parallel \frac{1}{2}(p(x)+q(x)) \right) + \text{KL} \left(q(x) \parallel \frac{1}{2}(p(x)+q(x)) \right) + \\
& + 2 \log \frac{1}{2} = 2 \text{JS}(p \parallel q) + 2 \log \frac{1}{2} \Rightarrow q_{\text{opt}}(x) = \\
& \quad \text{Jensen-Shannon divergence} \quad = p_{\text{data}}(x) \\
& \text{JS}(p \parallel q) \geq 0 \text{ and } = 0 \Leftrightarrow p = q
\end{aligned}$$



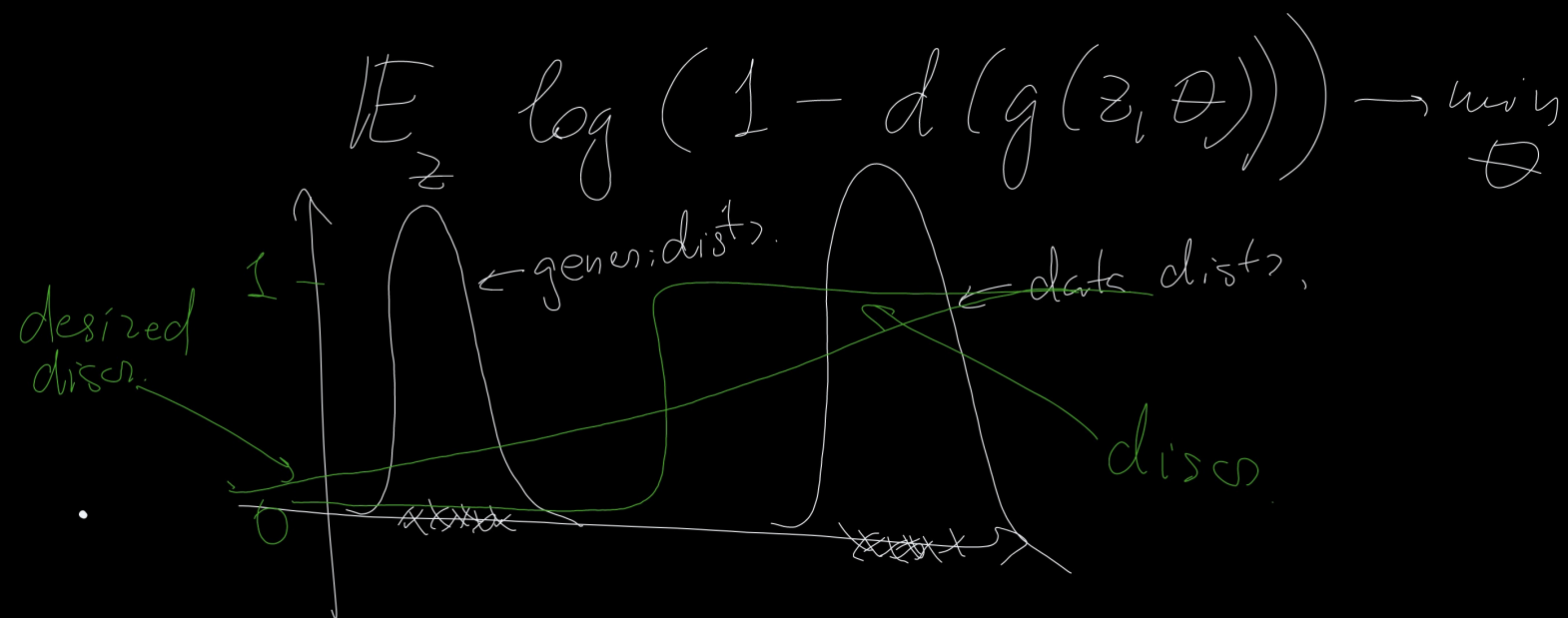
$$KL(p \parallel q) = \int p(x) \log \frac{p(x)}{q(x)} dx =$$

= not-defined

$$JS(p \parallel q) = \int_a^{a+1} 1 \log \frac{1}{\frac{1}{2}} dx +$$

$$+ \int_b^{b+1} 1 \log \frac{1}{\frac{1}{2}} dx = -2 \log \frac{1}{2}$$

Classic GAN training is unstable!

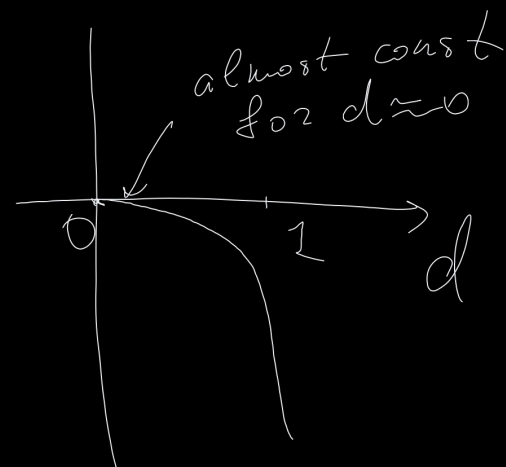


Tricks

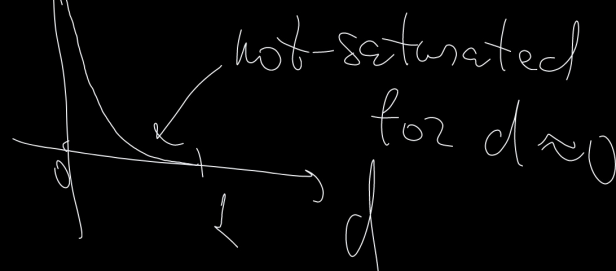
① Non-saturated loss for generator

$$\mathbb{E} \log(1 - d(g(z)))$$

$$d(g(z)) \approx 0$$



loss: $-\mathbb{E}_z \log d(g(z))$



② label smoothing
real 1 $\begin{cases} \rightarrow 0.9 \text{ real} \\ \rightarrow 0.1 \text{ fake} \end{cases}$

loss for disc: $E_{x \sim p_{\text{data}}(x)} (0.9 \log d(x) +$
 $+ 0.1 \log(1 - d(x))) + E_{z \sim p(z)} \log(1 - d(g(z)))$
 $\rightarrow \text{map } d$

③ Instance noise injection

$$x_1, \dots, x_M \sim p_{\text{data}}(x)$$

$$\varepsilon_1, \dots, \varepsilon_M \sim \text{noise}$$

$$z_1, \dots, z_M \sim p(z)$$

$$\tilde{\varepsilon}_1, \dots, \tilde{\varepsilon}_M \sim \text{noise}$$

$$- \frac{1}{M} \sum_{j=1}^M \left(\log d(x_j + \varepsilon_j) + \log(1 - d(g(z_j) + \tilde{\varepsilon}_j)) \right)$$

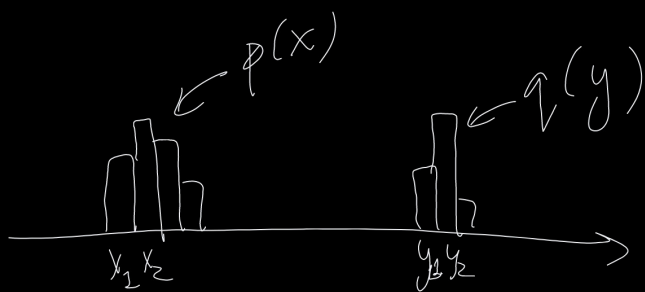
Least Squares GAN

$$d(x) \in \mathbb{R} \quad (\text{not } \in (0, 1))$$

$$\text{loss for disc.} : \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}(x)} (d(x) - 1)^2 +$$

$$\text{loss for generator: } \frac{1}{2} \mathbb{E}_{z \sim p(z)} (d(g(z)) - 1)^2 \rightarrow \min_g + \frac{1}{2} \mathbb{E}_{z \sim p(z)} (d(g(z)) - 0)^2 \rightarrow \min_d$$

Wasserstein GAN



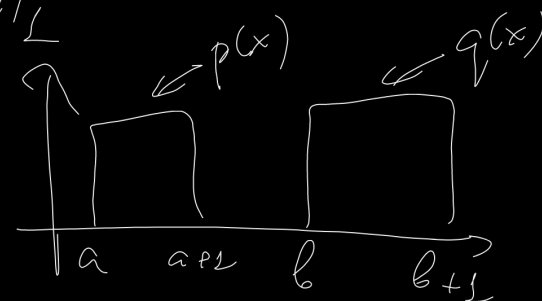
$\angle P \rightarrow$

transportation plan $\gamma(x_i, y_j)$
 transportation cost: $\|x_i - y_j\|_1$

$$\begin{cases} \sum_{x_i, y_j} \gamma(x_i, y_j) \|x_i - y_j\|_1 \rightarrow \min_{\gamma} \\ \gamma(x_i, y_j) \geq 0 \quad \forall i, j \\ \sum_{x_i} \gamma(x_i, y_j) = q(y_j) \quad \forall j \\ \sum_{y_j} \gamma(x_i, y_j) = p(x_i) \quad \forall i \end{cases}$$

$$W(p, q) = \inf_{\gamma \in \Pi(p, q)} \int \gamma(x, y) \|x - y\|_L$$

couplings



$$\gamma \in \Pi(p, q) \Leftrightarrow \gamma(x, y) \geq 0 \quad \forall x, y \quad W(p, q) = |b - a|$$

$$\int \gamma(x, y) dx = q(y) \quad \forall y$$

$$\int \gamma(x, y) dy = p(x) \quad \forall x$$

$$W(p_{\text{data}}(x), p_{\text{gen}}(x)) \rightarrow \min_{p_{\text{gen}}}$$

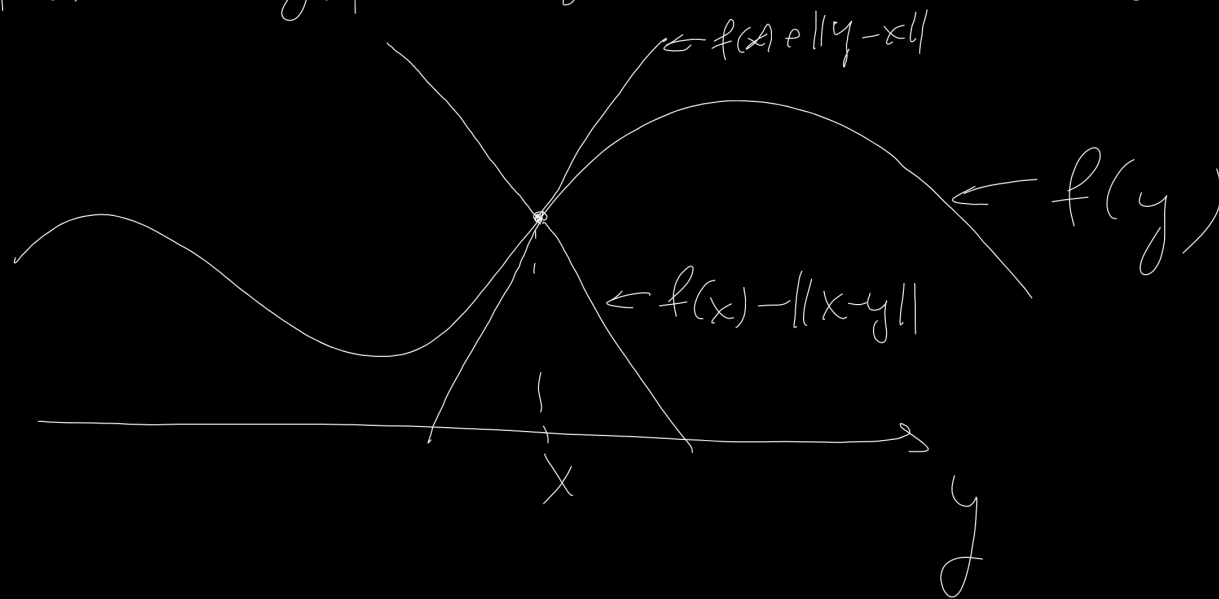
Th (Kantorovich - Rubinstein duality)

$$W(p, q) = \sup_{f \in 1\text{-Lipschitz}} \left(\mathbb{E}_{x \sim p(x)} f(x) - \mathbb{E}_{x \sim q(x)} f(x) \right)$$

$$f \in 1\text{-Lipschitz} \Leftrightarrow \|f(x) - f(y)\| \leq \|x - y\| \quad \forall x, y$$

$$\|f(x) - f(y)\| \leq \|x - y\| \stackrel{f \in C^1}{\iff} \|\nabla f(x)\| \leq 1 \quad \forall x$$

$$f(x) - \|x - y\| \leq f(y) \leq f(x) + \|x - y\|$$



$g(z, \theta_g)$ - generator

$f(x, \theta_f)$ - critic
 $\theta_f \in \mathbb{R}$

$$W(p_{\text{data}}, p_{\text{gen}}) = \sup_{\theta_f} \mathbb{E}_{x \sim p_{\text{data}}} f(x, \theta_f) - \mathbb{E}_{z \sim p(z)} f(g(z, \theta_g), \theta_f)$$

Loss for critic: $-\mathbb{E}_{x \sim p_{\text{data}}} f(x) + \mathbb{E}_{z \sim p(z)} f(g(z)) \rightarrow \min_{\theta_f}$ $\rightarrow \min_{\theta_g}$

Loss for generator: $-\mathbb{E}_{z \sim p(z)} f(g(z)) \rightarrow \min_g$

Gradient penalty regularizer

$$\text{loss for critic: } -\mathbb{E}_{x \sim p_{\text{data}}} f(x) + \mathbb{E}_{z \sim p(z)} f(g(z)) + \\ + \lambda \mathbb{E}_{x \sim \text{perturbed}} \left(\|\nabla_x f(x)\| - 1 \right)^2$$

$$x \sim p_{\text{data}}(x), z \sim p(z), \varepsilon \sim \mathcal{U}[0, 1]$$

$$x_{\text{perturbed}} = \varepsilon x + (1 - \varepsilon) g(z)$$

Spectral regularization

$$f(x) = Wx \quad \|f(x) - f(y)\|_2 = \|W(x-y)\|_2 \leq \|W\|_2 \|x-y\|_2$$

If $\|W\|_2 \leq 1 \Rightarrow f \in 1\text{-Lipschitz}$

$$\|h(f(x)) - h(f(y))\| \stackrel{h \in 1\text{-Lipschitz}}{\leq} \|f(x) - f(y)\| \stackrel{f \in 1\text{-Lipschitz}}{\leq} \|x - y\|$$

$$\|W\|_2 = \sup_{x \neq 0} \frac{\|Wx\|_2}{\|x\|_2} = \sigma_{\max}(W) = \sqrt{\lambda_{\max}(WW^T)}$$

$$W \rightarrow W_{SR} = \frac{W}{G_{\max}(W)}$$

$$\|W_{SR}\|_2 \leq 1$$

← assumed to be constant
in less backprop

$$A \in \mathbb{R}^{n \times n}, A = A^T \geq 0, A q_{\max} = \lambda_{\max} q_{\max}$$

Power iteration: $u_0 \sim \text{noise}$, $u_1 = \frac{A u_0}{\|A u_0\|}$, $u_2 = \frac{A u_1}{\|A u_1\|}, \dots, u_k \rightarrow q_{\max}$

$$A^K = Q \Lambda^K Q^T \quad A^K u_0 = \sum_{i=1}^n \lambda_i^K q_i (q_i^T u_0) \underset{\substack{\approx \text{const } q_{\max}^K \\ \lambda_{\max}^K \gg \lambda_{\text{second max}}^K}}{\approx \text{const } q_{\max}^K} \gg \dots \gg \lambda_{\min}^K$$

$$u_0 \sim \text{noise}$$

$$v = W^T u_0$$

$$v = \frac{v}{\|v\|}$$

$$u = Wv$$

$$u = \frac{u}{\|u\|}$$

$$\sigma_{\max}(W) \approx u^T W v$$